Unused descriptors versus DFT targets: Spearman rho
Common training cohort; pairwise complete observations

	OH energy	O energy	OOH energy	DFT overpotential
Adsorption metal: mean ligand dipole magnitude	-0.49	-0.50	-0.50	+0.29
Adsorption metal: spread ligand dipole magnitude	+0.19	+0.22	+0.23	-0.27
Adsorption metal: minimum ligand dipole magnitude	-0.44	-0.46	-0.45	+0.27
Adsorption metal: maximum ligand dipole magnitude	-0.67	-0.68	-0.67	+0.21
Adsorption metal: total ligand charge	+0.55	+0.57	+0.55	-0.22
Adsorption metal: mean ligand charge	+0.55	+0.57	+0.55	-0.22
Adsorption metal: spread ligand charge	+0.24	+0.25	+0.30	-0.31
Adsorption metal: minimum ligand charge	+0.70	+0.70	+0.67	-0.12
Adsorption metal: maximum ligand charge	+0.51	+0.54	+0.54	-0.24
Adsorption metal: total ligand spin magnitude	-0.13	-0.12	-0.13	+0.12
Partner metal: mean ligand dipole magnitude	-0.32	-0.35	-0.36	+0.21
Partner metal: spread ligand dipole magnitude	+0.35	+0.37	+0.40	-0.29
Partner metal: minimum ligand dipole magnitude	-0.35	-0.38	-0.39	+0.23
Partner metal: maximum ligand dipole magnitude	-0.18	-0.22	-0.22	+0.07
Partner metal: net ligand dipole magnitude	+0.18	+0.21	+0.24	-0.31
Partner metal: total ligand charge	+0.39	+0.42	+0.43	-0.21
Partner metal: mean ligand charge	+0.39	+0.42	+0.43	-0.21
Partner metal: spread ligand charge	+0.40	+0.42	+0.43	-0.20
Partner metal: minimum ligand charge	+0.25	+0.29	+0.30	-0.12
Partner metal: maximum ligand charge	+0.43	+0.46	+0.47	-0.18
Partner metal: total ligand spin magnitude	+0.03	+0.01	+0.00	+0.02
Mean ligand angle at adsorption metal	-0.22	-0.21	-0.22	+0.11
Spread of ligand angle at adsorption metal	-0.21	-0.21	-0.21	+0.13
Smallest ligand angle at adsorption metal	-0.23	-0.23	-0.22	+0.02
Largest ligand angle at adsorption metal	-0.22	-0.21	-0.22	+0.11
Mean metal-bridge-metal angle	+0.32	+0.30	+0.29	-0.08
Spread of metal-bridge-metal angles	+0.43	+0.41	+0.34	+0.28
Mean bridge dipole magnitude asymmetry	-0.12	-0.12	-0.10	+0.15
Partner metal spin magnitude	-0.08	-0.10	-0.13	-0.05
Signed metal spin product	+0.36	+0.33	+0.36	+0.03
Net support spin	+0.19	+0.14	+0.11	+0.02
Total support spin magnitude	-0.24	-0.24	-0.27	+0.14
Ligand nonplanarity	NA	NA	NA	NA
Metal displacement from ligand plane	-0.35	-0.33	-0.35	+0.32
Shared ligand count	NA	NA	NA	NA
Terminal ligand count	NA	NA	NA	NA
Adsorption metal: mean shared ligand charge	+0.51	+0.52	+0.52	-0.20
Partner metal: mean shared ligand charge	+0.51	+0.52	+0.52	-0.20
Partner metal: mean terminal ligand charge	-0.01	+0.06	+0.08	-0.02
Second-shell carbon density near 2.8 Angstrom	+0.04	+0.03	+0.02	+0.25
Second-shell carbon density near 3.6 Angstrom	+0.06	+0.06	+0.08	-0.20
Second-shell N/O/S density near 2.8 Angstrom	-0.19	-0.17	-0.16	+0.07
Second-shell N/O/S density near 3.6 Angstrom	-0.19	-0.17	-0.16	+0.07
Mean ligand angle at partner metal	-0.31	-0.31	-0.31	+0.22
Spread of ligand angle at partner metal	-0.28	-0.28	-0.29	+0.22
Smallest ligand angle at partner metal	-0.36	-0.35	-0.34	+0.19
Largest ligand angle at partner metal	-0.31	-0.31	-0.31	+0.22

